

**How do government subsidies for clean energy affect household
energy choices in low-income regions?**

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Abstract:

As the world is nearing the end of the 2nd decade of the 21st century, the world at large has recognized (or is starting to realize) the need and significance of clean energy. Whether it is in international climate conferences, local government policies, emergence of clean alternatives to fossil fuels or the spread of the discourse around climate change among the masses, it is clear that clean and green energy is the future. This research paper discusses a very vital element of this climate revolution, government subsidies and their role in making clean fuel a household thing even in low income groups. The paper will discuss key concepts, major policy decisions and recent developments in the area in a span of 2 well organized chapters and in the last chapter, will conclude the study by discussing pertaining issues and possible solutions.

Keywords: Clean energy, Subsidies, United Nations, India, Climate Change, Global Warming, Low-income groups.

Context and Introduction

In the advent of the current global energy crises, governments worldwide have strived to implement effective policies and commit funds to a solution. Since the start of global energy crises, governments have spent about 900 billion dollars on quick fixes to cover energy costs for consumers, alongside pre-existing support programmes and subsidies.

In addition, it is evident that socio-economic inequalities play a major role in the policy making around clean energy reforms. Research findings urge governments to take into account targeted and complementary policies to enhance outcomes as well as identify economically vulnerable groups whilst designing effective strategies and programmes. For example, the

Brazilian Energy Office had developed the Electrical Gini Index to measure inequality in residential electricity use across income groups. The Electrical Gini Index serves as a tool to gather data on the ownership and power usage patterns of home appliances across different income groups and regions.

It may come as a surprise but a common, majority consensus that climate change is real and needs to be battled is a very recent phenomenon. Although the number of people (especially the younger generations) who believe that climate change is not real and can be ignored has now drastically reduced, there have always been lobbies and people in the society who have actively battled the initiatives on climate change and are of the opinion that climate change is but a propaganda. Unfortunately, some of these people are not nobodies and hold a significant influence.

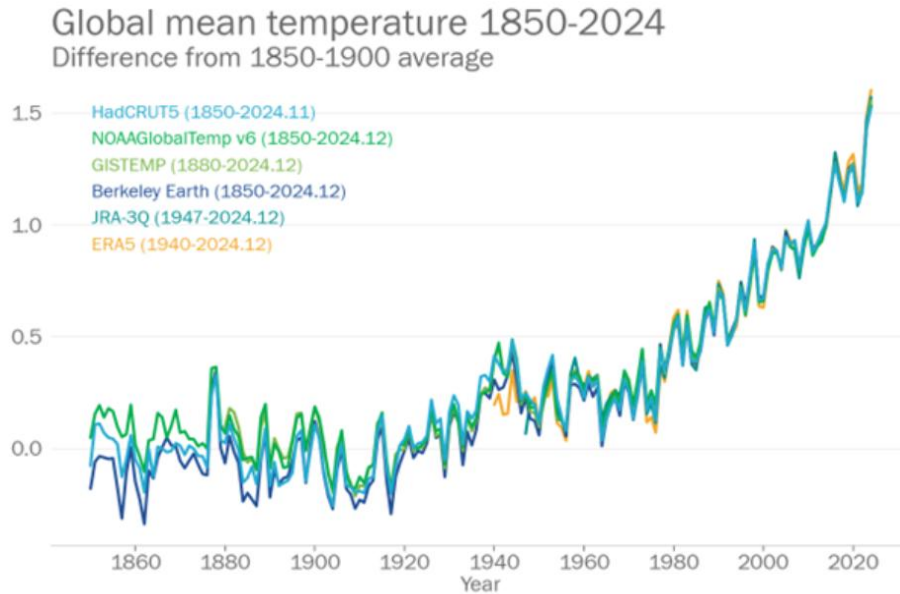
Apart from the greedy and capitalist fossil-fuel based companies that deny climate change just like tobacco companies used to deny a link between tobacco and cancer (Goldenberg, 2015), many important names in world politics also share this stage. In 2010, Donald Trump, the current president of the USA, said that the world is now getting ‘colder’ and that the noble laureate Al Gore, recognized for his role in disseminating information about climate change, is a hoax and his noble prize should be taken back (Schulman, 2018). He even believed in 2012 that the discourse around climate change was an ‘invention’ of China to take dominance over America in manufacturing industries (Phillips, 2016). He went on further in early 2020 when he and his allies entirely dismissed climate change and even said that there should be no studies at all on this subject (Roston and Kahn, 2025). He is however, not the only climate change denier. Many US senators like Jim Inhofe, Tom Coburn, Mo Brook, Majorie Taylor Greene and many more have either made uninformed claims or absolutely denied the

sensitive subject (Cama, 2015) (Brown, 2013) (Waldman, 2018). Statistically, 23% of the members in the 118th congress of US senate ardently denied climate change (So, 2024), which is a very worrying stat for common masses. Brazil is another instance that has baffled climate change scientists because in 2019, the country's minister of foreign affairs under a far right Bolsonaro government declared that climate change is a 'marxist' propaganda and even eliminated Brazil's climate change related ministries (Watts, 2018) (Escobar, 2019).

More research has been conducted in Northwest China, a low-income region, to examine the factors influencing the choices and adoption of clean energy in rural households. In recent years, research on rural energy consumption shows that the main fuels used fall into two groups namely solid fuels (e.g. straw fuelwood and coal) and clean fuels (biogas, natural gas, liquified gasoline and electricity). In some areas, they found that kerosene remains the primary cooking fuel, accounting for approximately 49.1% of households. Additionally, research found that coal consumption decreases by approximately 17% for every 10% increase in rural household income, reflecting a gradual shift toward modern energy sources.

The reality is however, significantly different. Global temperatures are reaching an all-time high and according to the World Meteorological Organization, 2024 was registered to be the hottest year on record, with global average surface temperature being 1.55 degree celsius above the 1850-1900 average. Even though this does not mean that long term climate goals set by the Paris Agreement¹ are in immediate danger, but even a fraction of change in global temperature needs to be rectified to avoid disastrous consequences in future (WMO, 2025).

¹ To be discussed in later in the chapter



(WMO, 2025)

Owing to the vast scope of research that has been done and is going on about climate change, it has been established that it is one of the leading causes of natural disasters around the world (USGS, 2025). Many people often confuse climate change with global warming. While both terms are interrelated, global warming is a part of climate change and refers primarily to the increasing temperatures around the world. Climate change is a wider concept that includes melting of icebergs, decrease in forest cover, rising temperatures, irregular and untimely precipitation and so on. Some of the long term hazards of climate change include persistent droughts in some arid areas, perennial floods in some coastal regions, decrease in snow and forest covers and subsequent destruction of flora and fauna, the rise in sea levels, shore-line erosions and occurrence of natural disasters. Although these disasters are often unalarmed and happen occasionally, this does not mean that they are to be given less attention than the long term impact. The following paragraph will elaborate on some of these natural disasters and explain why they demand urgent catering to.

For as long as there have been life forms on the planet, there has been climate and hence, natural climate disasters. However, the involvement of anthropogenic factors (human activities that lead to climate change) has exponentially increased the appearance of these disasters. Some of these anthropogenic factors include burning of fossil fuels like coal and oil, deforestation, over exhaustion of plant and animal based products, industrial waste generation, unchecked infrastructural development and many such activities that harm the environment. Many international organizations research extensively on the subject and have tried to unearth the increase in occurrences of natural disasters over the years. One of the major available datasets on this subject is released by the International Disaster Database (EM-DAT) of the Centre for Research on the Epidemiology of Disasters (CRED). It is noteworthy that many organizations of international significance like the United Nations, World Meteorological Organization and the United Nations Framework Convention on Climate Change (UNFCCC) also use this data. Although the data is transparent, open source and well researched, the dynamic space of climate research is such that there can be variations or discrepancies in it due to some unreported or hard to quantify disasters. Based on multiple such ascertainties and these analyses, the Food and Agricultural Organization (FAO) in 2023 said that climatic disasters have nearly quadrupled from averaging 100 per year in the 1970s to over 400 per year in the 21st century. The claim was then also backed by the UN chief. The UNFCCC and WMO estimate this increase to be 5 times. The legitimacy of these claims can be confirmed through their repeated citations in leading news outlets like BBC and The Economist (Ritchie and Rosado, 2024). The increase can be 4 times, 5 times or even 3 times but for sure, it has been registered and needs rectification before the numbers go further up. Given below is a line graph that depicts the steep rise in occurrence of disasters from the year 1900 to April 2024.



(Ritchie and Rosado, 2024)

If one tries to ascertain the practical effects of climate change, he or she will soon discover that it ranges from financial loss to the loss of life and increasing the wealth disparity. All around the world, people are losing their families and homes to natural disasters more than ever. When the steep increase in population in many parts of the world is paired with more than ever exploitation of natural resources and a low rate of their replenishment, it becomes clear that it is the lower classes of the society that suffer the worst. Some more frightening data suggests that just between the 10 years from 2006 to 2016, the rise in sea levels across the world was 2.5 times more than it was in the entirety of the 20th century. Over 20 million people every year are forced to evacuate their houses due to climate change and as per United Nations Environment Programme's estimate, the rehabilitation and adaptive cost of the havoc caused by these disasters in developing and under-developed countries is 140-300 billion dollars. Some of the deadliest examples of this include the chronic and widespread droughts in East African countries like Ethiopia, Somalia and Kenya that have wiped off crops, killed livestock forced communities to migrate, the annual South Asian floods in many parts of countries like India, Bangladesh and

Nepal which cause economic disruptions and loss of life on a large scale and existence of a dry corridor in many agriculture depended countries like Honduras, El Salvadore and Nicaragua in Latin America due to the dry months season growing from 3 months to 6 or more months (Oxfam, 2025). However, mother nature does not spare developed countries from this destruction either. Many developed and first world countries regularly suffer from hurricanes, floods and wildfires.

To illustrate through example, the Australian wildfires of 2020 were a follow up from the country's hottest ever year and worked their way to destruction through the dry and waterless soil. For record, the fire was so widespread that it burnt over 10 million hectares of forests to the ground and claimed the lives of a billion of native fauna species, a loss the eco-system might never recover from (Oxfam, 2025). A more recent example is of the Los Angeles fires of 2025 that accounted for a total loss of staggering 250 billion dollars (Vincent, 2025). The extent of this disaster was so huge that it is being called one of the most terrible natural tragedies in US history. The fires wiped off thousands of acres of forest covers and forced well over 200,000 people to migrate from their houses. The scientists have estimated reduced precipitation, dried-out forest covers and persistent droughts to be the reason for this massive destruction and continue to warn that if the burning of fossil fuels does not stop, the states like California are likely to experience these fires again, probably with a more destructive force. In the words of Dr, Friederike Otto who is the co-lead of World Weather Attribution and Senior Lecturer in Climate Science at the Imperial College, London:

“Climate change is continuing to destroy lives and livelihoods in the US. From violent hurricanes in the east, to nightmarish wildfires in the west, Americans are experiencing the devastating consequences of fossil fuel warming. In 2025, the choices facing world leaders

remain the same – to drill and continue to burn oil, gas and coal and experience ever more dangerous weather, or transition to renewable energy for a safer and fairer world” (Levey and Frazer-Baxter, 2025).

The world certainly does not need any more reminders to accept climate change and switch to cleaner fuels while there is still time.

What is Clean Energy and Why is it Important?

Clean energy refers to the fuels or energy sources that do not contribute to global warming because they do not emit or limit the production of greenhouse gases like carbon dioxide, methane, nitrous oxide or other air pollutants. Clean energy is often taken to be synonymous with green energy but while green energy comes entirely from natural resources, clean energy is a wider term that may include both renewable and non-renewable. For instance, while nuclear energy is clean and does not harm the environment, it is generated from Uranium, a finite resource. Wind energy, solar energy, hydroelectricity, biomass, tidal energy and geothermal energy are some of the major sources of clean energy.

There are many advantages to using clean energy. Some of the major benefits of clean energy for the global community can be that first of all, most of these energy sources are inexhaustible, meaning that not only do they not harm the environment, they are also never going to stop providing, contributing to the sustainable industrial and domestic usability, energy sources like wind and solar energies can also help in making countries financially independent in a way that they do not have to rely on bigger and richer countries for their domestic energy needs. There are also many socio-economic benefits of these sources of energy. Clean energy is also one of the most rapidly growing and still developing areas of study and so, it encompasses

in it, the opportunity of creation of new jobs that can boost the economic premise of the countries. Moreover, since these energy sources do not harm air or water, they are also advised over fossil fuels to protect the health concerns of citizens.

Realizing the positive implications of clean energy for a better future of mankind and the urgent requirement to make it a household name by decarbonizing the energy usage of the upcoming generations, the UN General Assembly has also declared January 26 as the ‘International Day for Clean Energy’ (Iberdrola, 2025). As a matter of fact, ‘Ensuring access to affordable, reliable, sustainable and modern energy for all’ has been declared as one of the 17 sustainable development goals adopted by the United Nations, to be fulfilled by 2030 across all recognized nations in the world (UNO, 2025a). Although ‘Ensuring Environmental Sustainability’ was a Millennium Development Goal (to be fulfilled in the years between 2000 and 2015) (Dag Hammarskjöld Library, 2025a), there was never the urgent need to adopt cleaner energy sources such widely expressed as done through the SDG 7.

Global Outlook Towards Clean Energy

For the common people around the world to know about and understand climate change, the efforts to spread awareness, research and practical rectifications could only begin with international organizations or governing bodies taking action. First step in this direction came through the ‘Ramsar Convention held in Ramsar, Iran in 1971 also known as the Convention on Wetlands of International Importance. Till date, it is the only global multilateral environmental agreement that focuses on a single ecosystem (Ramsar, 2015). The United Nations’ first climate conference was held in Stockholm in 1972 after the Economic and Social Council recommended the United Nations General Assembly to convene a programme on the issues of Human Environment. The major significance held by the Stockholm Conference was that it led to the

formation of the United Nations Environment Programme (UNEP). This was followed by the formation of the World Commission on Environment and Development in 1987 and based on a four year study, the commission submitted its report titled 'Our Common Future'. This was the first time the term 'Sustainable Development' found its way to the forefront of discussions (Dag Hammarskjöld Library, 2025b). Since then, various United Nations' environment oriented bodies have been formed and multiple summits, conferences and treaties have been signed. The following paragraph will talk about some of these major events.

One of the earliest and landmark treaties that happened in the early days of climate change action was the Montreal protocol of 1987. It sought to regulate and monitor the use and production of over 100 man-made chemicals that damaged the Ozone layer (UNEP, 2025). The Basel Convention on the Control of Transboundary Movements of Hazardous Wastes and their Disposal is another United Nations treaty that was signed in 1989 and came into effect 1992 (Basel Convention, 2025). A major summit that followed this was the Rio summit in 1992, also called the United Nations Conference on Environment and Development or the 'Earth Summit'. It was in this summit that the Rio Declaration on Environment and Development was signed which presented a series of principles and the Agenda 21 was declared, which was a global plan dedicated to sustainable development (Dag Hammarskjöld Library, 2025b). Adapted in Kyoto, Japan in December of 1997 was the Kyoto Protocol, another landmark event in the history of climate change. The Kyoto Protocol operationalizes the United Nations Framework Convention on Climate Change (UNFCCC) wherein basically, it commits manufacturing countries and developing economies to control and monitor greenhouse gases (GHG) emissions in accordance with their own personally set and agreed targets. The Convention, however, asks these countries to adopt policies based on the protocol and to report to it periodically (UNFCCC, 2025a).

Although there have been many many more such important summits and conferences on the subjects related to climate change, sustainable development and global warming, mentioning all of them here is not significant to the research subject at hand. The following paragraph will discuss a few more of these conferences that go hand in hand with this research.

In December 2015, members of 195 participating parties met in Paris at the UN Climate Change Conference (COP21). They signed what came to be known as the Paris Agreement. The agreement entered into operation a year later, in November 2016. The main aim of this conference was to limit “the increase in the global average temperature to well below 2°C above pre-industrial levels” and make sure to keep “the temperature increase to 1.5°C above pre-industrial levels” (UNFCCC, 2025b). In a more recent development, the secretary general of the United Nations, Antonio Guterres addressed and reported on "A Moment of Opportunity: Supercharging the New Energy Era" on July 22, 2025. He emphasized on the inevitable and beneficial shift towards clean and renewable energy and presented a brief outline of the procedure. To elucidate in his words, “The fossil fuel age is flailing and failing. We are in the dawn of a new energy era” (UNO, 2025b).

India’s Major Official Policies Pertaining to the Climate Change and Clean Energy

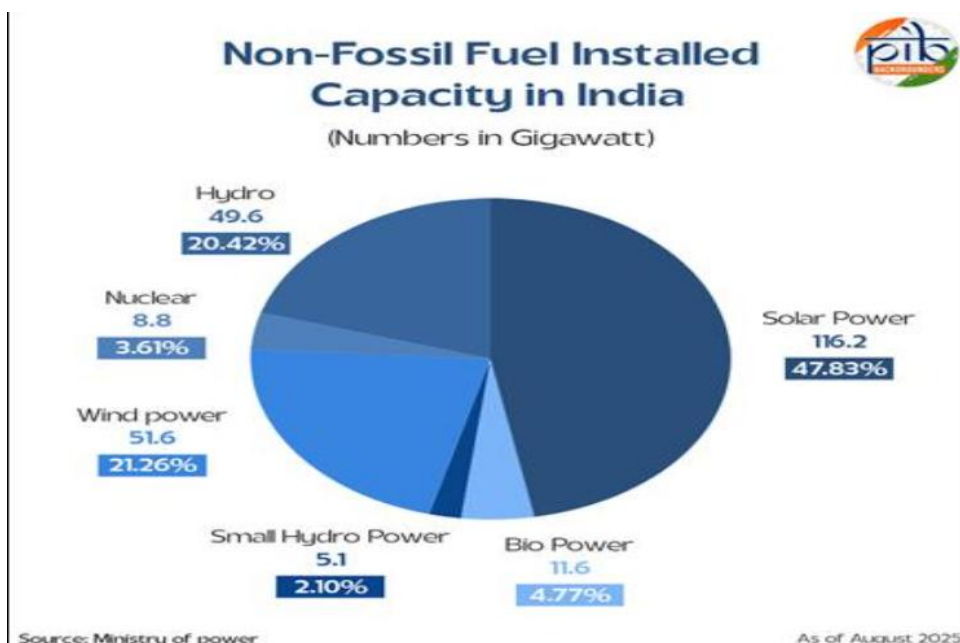
The government of India has always been adherent to the environmental studies, protocols and targets set by international organizations like the United Nations and continuously works towards achieving them. Just over a decade ago, seeing solar panels on the rooftops of houses was a rare sight and only showed itself in big cities. As of 2025, India ranks 4th in the world in terms of renewable energy installed capacity of wind energy and 3rd when it comes to solar energy. As per the pledge taken under COP26, India will have to achieve generating 500 GW of non-fossil fuel electricity capacity by 2030 (PIB, 2025).

India has made notable progress in the area of expanding solar energy. As of 2025, India's combined solar energy capacity stands at 119.02 GW. The total solar GW capacity of India is 748 GW and efforts are being directed to its achievement. Another beneficial scheme that connects subsidy and solar energy is the PM KUSUM (Pradhan Mantri Kisan Urja Suraksha Evam Utthaan Mahabhiyaan) scheme which aims at promoting the use of solar energy instead of diesel among the farmers. Under this scheme, farmers can get a 30% to 50% subsidy to install new solar pumps or convert old pumps to solar. They may also build solar power plants which generate up to 2 MW energy on their farmlands and sell them to local DISCOMs to earn profit. This scheme targets at ensuring energy security to farmers and also assists India's goal to replace at least 50% of fossil fuels with clean energy by 2030. Under the "Development of Solar Parks and Ultra Mega Solar Power Projects" scheme, the government has approved 53 solar parks across 13 states with a total capacity of 39,323 MW. Moreover, floating solar parks like the Omkareshwar Floating Solar Park in Madhya Pradesh have been approved to save land for other purposes and generate more energy efficiently. Although there are various other big and small developments in the area of solar energy set up in India, this paragraph will conclude with discussing two of the major ones, International Solar Alliance (ISA) and One Sun, One World, One Grid (OSOWOG) (PIB, 2025).

ISA was launched by India and France in 2015 at COP21. It is an alliance of 100+ nations where the main objective is to battle climate change through solar energy. The alliance aims at generating 1 trillion dollars by 2030. By bringing together developed, developing and under-developing countries, ISA aims to work towards a carbon emission free and optimal alternative to burning fossil fuels. Likewise, OSOWOG too works towards the same objective

by aiming to connect solar resources across the regions of the world. It was a joint initiative of India and ISA launched in 2018 and hoisted the slogan “the sun never sets” (PIB, 2025).

Given below is a pie chart depicting the share of different clean energy fuels in the total installed capacity of India.



(PIB, 2025)

Another giant leap in the direction of clean energy by India is cutting the subsidies on fossil fuels. Guided by the bold target of zero net carbon emission by 2070. According to the Asian Development Bank (ADB), India has cut its fossil fuel subsidies by 85% already under its ‘remove, target and shift’ strategy from 25 billion dollars in 2013 to 3.5 billion dollar 2023. Petrol and diesel have been put through the gradual phasing out strategy by increase in excise duties and boosting tax revenues. Likewise, the government from 2010 to 2017 implemented a cess on coal production and imports. What happened as a result was that around 30% of collections from this were allocated to the national clean energy and environment fund (PIB, 2024).

In the fiscal year 2023, India's energy subsidies were recorded at 39.3 billion dollars. This is largely attributed to the nation's well thought strategy and planning about its climate change (Anand, 2024). In fact, balancing growth and commitments towards battling climate change was also the key stance of India in COP28 and COP21 and thus far, the country has shown tremendous progress. For instance, under UNFCCC and Paris Agreement, India had officially submitted its first Nationally Determined Contribution (NDC) in 2015 which the two major quantifiable targets were to reduce the emission intensity of its GDP by 33% - 35% by 2030 from 2005 and to achieve 40% clean energy power installed capacity from non-fossil fuel based energy resources by 2030. India achieved these targets way before time scheduled. As of October 2023, the cumulative electric power installed capacity of India was 186.46 MW, making it almost 44% of the total installed capacity (4% over the desired capacity). Similarly, the emission intensity of its GDP has been reduced by 33% between 2005 to 2025. India updated its NDC in 2022. Now, the new target for reducing emission intensity of India's GDP to be achieved is 45% by 2030 from 2005 levels and installed capacity from non-fossil fuel-based energy resources has been enhanced to 50% by 2030 (PIB, 2023).

This far, India's commitment to clean energy adoption has been unfettered and relentless. This progress has definitely fostered people's trust in the government's intentions towards battling climate change and this will in turn make more and more people shift to clean energy adoptions.

What are Government Subsidies and How They Work?

In simple words, subsidies are the financial benefits provided by the governments, sometimes by direct payments like cash or indirect means like tax reductions. The basic idea behind subsidies is public welfare through reducing the financial burden on people. They can

sometimes be used as means to promote new economic public policies. Subsidies often come with an opportunity cost. For instance, a subsidy provided to the farmers may ensure that they are able to avail the latest technology and hire more workers for increased production but on the other hand, since subsidies are given from taxes, they might lead to an increase in price of groceries for the customers (Investopedia, 2025).

However, a debate has always existed on the issue of subsidies. While it is advisable for the developing countries to provide them in various sectors of economy like agriculture, energy and real estate to help them boom in their infancy, one group of free market proponents argue that this may disproportionate the economy and that technically, any business that cannot survive the market forces should not be revived on the cost of general public's taxes. Moreover, it is also possible that the benefits of subsidies might just not reach the actual needy. Regardless of the view anyone holds, it is important to understand that in countries like India that are a welfare state, subsidies are natural to exist. In fact, researches have shown that subsidies on the means of agricultural production have actually contributed to increased production, food security and constance in price of commodities in the long run (Seemon, 2025).

In the context of the subject matter at hand, providing subsidies on clean energy sources can definitely lead to not just increased everyday use of renewable energy but also, financial benefits to the farmers, other consumers and even the producers of the renewable energy. In a study conducted in Jiang Xi, China on sustainable bioenergy development and government subsidies, it was found that lack of awareness was not the reason farmers abstained from using bio energy sources over fossil fuels; it was the cost associated with it and hence, as the cost was lowered, the usability increased. It was also found that using ethanol, which releases a significantly low amount of greenhouse gases, could potentially replace over 400 million tonnes

of gasoline used every year and potentially took the farmers' earning per hectare from 11,136 dollars to 15,121 dollars. This subsidy, however, cost the government somewhere between 225 million and 246 million dollars annually but on the flip side, helped farmers and ethanol producers in gaining around 800 million to nearly 2000 million dollars annually. In terms of environmental benefits, using ethanol with bio-electricity (generated through pyrolysis which is a process under which agricultural waste is heated in absence of oxygen to produce electricity) offset up to 1.28 million tonnes of carbon dioxide every year. Moreover, 'biochar', a by-product of pyrolysis can enhance soil fertility significantly and this process, in long term, may entirely eliminate carbon production the world is aiming at (Kung, Zhang and Kong, 2016).

An example of subsidy on energy from India can be of the PM Surya Ghar: Muft Bijli Yojana (free electricity scheme). Under this scheme, the mission is to provide 300 units of free energy per month to over 1 crore houses in India with a combined budget of 75,000 crore. The scheme comes packed with subsidy induced benefits to both the residential households and Resident Welfare Association/ Group Housing Societies. For residential households, Rs 30,000 is allotted per kW of energy use, up to 2 kW, Rs 18,000 for per kW for additional capacity up to 3 kW and Rs 78,000, total subsidy for systems larger than 3 kW capped at. For the residential societies, Rs 18,000 per kW is assigned for common facilities like EV charging, up to 500 kW with a few more conditions. For the special states, an additional 10% subsidy is applicable per kW (MNRE, 2025). The reason what makes subsidies a working tool for combating climate change and attracting common people to using it is the monetary gain associated with it and with policies like this, the government skilfully sets free the untapped potential of the country in terms of the future of energy, renewable and clean energy.

Low Income Groups and Energy Consumption

Generally, the observed pattern suggests that when it comes to the low income groups, the choice of energy source is majorly dependent on the cost of it. Many studies (as the ones that will be discussed here) have pointed out that mere awareness about the hazards of climate change will not be able to steer these people towards green and clean energy unless these sources can be attainable and affordable to them. This is where the need for government given subsidies come into picture. If we as a society want everyone, including the lower classes to reduce their carbon emission, we as a society should also make sure that these sources of energy do not burn holes in these people's pockets.

An important term that is noteworthy at this juncture is 'energy poverty'. In everyday context, energy poverty refers to the lack of access to modern and less carbon emitting sources of energy (Bouzarovski and Petrova, 2015). Recorded data suggests that as many as 2.6 billion people in the world, out of which 800 million or more reside in India, suffer from energy poverty (Clarke and Sharma, 2015). These people form over 1/5th of the global population and it goes without saying that if the governments around the world are able to shift them to cleaner energy sources through a planned and structurally organized procedure, the decrease in carbon emission will be significant. Currently, most of these people have to rely on wood, coal and multiple such pollution causing fuels for their daily kitchen needs. Aside from harming the environment, the fumes from these fuels can be hazardous to the indoor air and claim over 1.5 million premature deaths in India alone (Clarke and Sharma, 2015). Another aspect of this is that a majority of these people reside in developed or developing nations where women are the primary victims of this polluted air as they are made to work in the kitchen, not men. Provision of LPG or other cleaner fuels to these low earning families can free women from this shackle of breathing

hazardous air. Although the subsidy on LPG gas cylinders has been active in India since 2014 (MoPNG, 2023), many people are still devoid of it. To support the claim that higher income levels are directly proportional to a household's shift towards cleaner sources of energy, a study was conducted in Bangladesh (Haq et al., 2024) which studied the adoption of clean energy (dependent variable) in relation to various independent variables like rural-urban disparity, sex, education, internet and mobile ownership and connected to all of these, financial status. The study had some interesting findings which will be discussed in the next paragraph.

For context, Bangladesh has worked its way up in the 2010s from being a 'bottomless basket' to have consistently grown its economy with a 7% GDP growth rate in that decade. In that process, however, the country has heavily relied on fossil fuels to the fact that according to some datasets, in 2019, over two third of Carbon Dioxide emission in Bangladesh was a result of burning fossil fuels (Haq et al., 2024, p.2). Bangladesh, despite contributing less than 0.5% to the global greenhouse gases emission, ranked 7th in the global climate risk index of 2021 and is one of the 195 countries that have ratified the Paris Agreement of 2015 (p.2). Part of the reason why the country finds itself in such a deplorable state against battling climate change is because of extreme lack of awareness and carelessness from the governments. There has been no comprehensive pan nation study about what determines the energy choices of people and what are some of the variables to that. This study from Bangladesh derives its sample size and source from Bangladesh Multiple Indicator Cluster Survey. It consisted of 61,242 households across 8 divisions of Bangladesh and used two staged stratified cluster sampling. The sample had 22.1% population belonging to cities and 77.9% people from rural areas. 41.4% of the sample consisted of poor people, 19.4% people were from the middle class and 39.1% belonged to the upper or rich category (p.5). Almost everyone (94.8%) owned a mobile phone (p.7). Out of all these

categories of people, 80.1% of people did not adopt clean energy and only as few as 20% people used renewable and clean energy to fulfil their energy demands (p.5). Geographically, only the capital city division (Dhaka) had as many as 44.4% of the population who had adopted clean energy but there were some districts like Kurigram and Lalmonirhat where the clean energy adoption rate was 5% only (p.7). After assessing the data and analysing it through statistical models like Chi-square tests, EPRM (Extended Probit Regression), Concentration Index (CIX) and Blinder-Oaxca Decomposition, certain revelations came out. As it turned out, only 1/5th of Bangladeshi households used clean fuels. The rest of them relied on biomass for cooking and other indoor purposes. The adoption of clean energy in Bangladesh was 6 times higher in urban areas compared to rural areas. Moreover, higher levels of literacy implied an increase in clean energy adoption and so did higher income (pp.10-13). It can be concluded through this study that what Bangladesh needs is an organized campaign on climate change which would include properly linking the state policies with SDG 7 and provisions for right application and allotment of subsidies on clean energy fuels. It can be learnt through the example of Bangladesh that education, internet access and financial stability are some of the key areas to focus if any nation wishes to adopt clean energy to its maximum potential.

Finally, the last study this paper will look forward to for explaining the link between financial stability and adoption of clean energy is a study published in 2023. The primary research problem poised by the study was finding if the richer countries use more renewable and clean energy? A U shaped study was used which sought to verify the hypothesis: poor countries have less income and hence, less renewable energy usage and rich countries have more income and hence, more renewable energy usage. For this purpose, it looked at 8 Asian countries; India, Pakistan, China, Indonesia, Korea, Malaysia, Philippines and Thailand between 1990 to 2015

(Ergun and Rivas, 2023). This study was very interesting as it brought out some less discussed perspectives. The data it collected included REC% (share of clean energy in total energy matrix), GDP per person, Financial development and Capital formation and it showed that these countries, on an average, used 29% renewable energy (higher than the global average of 17.5%). By country classification, the highest number of users were in Pakistan (50%), India (48%) and Indonesia (45%) and the lowest number of users were in Korea (1%) and Malaysia (7%). The income levels ranged between very poor (890 dollars in Pakistan) to rich (17,356 dollars in Korea) (pp.5-6). A simple takeaway from this was that as countries got rich initially, they used more fossil fuels and less renewable sources but once they were rich enough, they would shift to renewable energy again. So the solution is for the governments to strive for energy and financial stability and this can be helped partially through subsidies.

Challenges and Way Forward

After having discussed various dynamics of clean energy on global and national level including its significance, importance, consequences if not taken seriously and the progress we have made thus far in this sector, it now also becomes very important to scan through some issues that have come up in the process. While there have been many brilliant policy decisions to propagate the use of clean energy, there have always been some on the other side of the spectrum. According to a recently published article, starting July 2025, any new solar or wind project which gets commissioned, is required by the law to pay 25% of the power transmission cost when it is across state borders. And this cost is expected to only increase over time. On an immediate level, this policy change may increase the tariffs by up to rupees 0.40 per kWh, which is almost a 16% hike. It is noteworthy that owing to irregular demand and high prices, about 30 GW of clean energy is already in crutches in India. Moreover, this policy of trans-border tariffs

will also lead to regionalism and decentralization and states may begin to favour renewable energy installments within the state boundaries, hampering inter state trade and affecting businesses involved. For instance, the state of Karnataka has already mentioned that they plan on installing the projects that only use transmission systems built within state boundaries to cut unnecessary cost on production (Bloomberg, 2025). The government should urgently look into such issues as they may hamper not just the spread of inter-state clean energy businesses but in the long run may also clash with the long term goals set by India.

Another issue that may hamper India's hopes to completely go carbon free by 2070 is the discrepancy in planning and execution. Apart from the rampant corruption and favoritism that cripple almost every other sector, there are actually issues specific to this subject. As discussed above, India achieved a major milestone when 5 years ahead of its set duration, the country was able to match the set target of establishing clean energy apparatus to fulfil 50% of entire energy demand. However, what gets missed when someone reads this headline is that the actual energy production has just increased from being 17% in 2014-15 to 28% by April 2025. This difference can be attributed to something called the 'Capacity utilization factor' (CUF) which means that renewable sources like wind and solar energy operate far below their capacity levels (25% to 30%) compared to the capacity of coal (60%) or nuclear energy (80%). Moreover, while solar and wind energy are largely dependent on external factors like wind speed and sunshine which are not always available, fossil fuels can be combusted on demand. This is the reason why coal is still used in abundance for power generation which according to some sources can even be 3/4th of the entire energy ecosystem of the country. Moreover, since India (and even the entire world) is still learning about the efficient ways to harness newer, cleaner forms of energy, there may arise issues related to storage and infrastructure of these sources of energy (Koshy, 2025).

In the World Population Report of 2024, The UNO has predicted that the global population will keep rising in the decades to come and will reach from 8.2 billion people currently to over 10 billion people in 2080s (Iberdrola, 2025). This clearly indicates that fossil fuels like petroleum and oil will not be able to quench the thirst of everyone, either due to unavailability caused by overuse or extremely high prices. Either way, it is inevitable for the world to switch to clean energy sources to not only meet the growing needs of the population but also prevent environmental hazards and natural calamities from occurring. The various conferences, summits and other awareness-cum-policy making programmes initiated around the world suggest the urgency shown by international organizations and gradually, people around the world. A good sign of this and a ray of hope is provided by the EIA (Energy Information Administration), USA which says that by 2050, the global energy demand will primarily be fulfilled by clean and green energy substitutes to fossil fuels (Iberdrola, 2025). This clearly means that an increase in human population and scarcity of fossil fuels will ultimately prompt governments around the world to shift to and hopefully, mandate the use of clean energy in the coming future.

For India and the majority of the world that is still figuring its way around clean energy, the best way forward is to celebrate how far they have come but also be prepared to face some setbacks if they arise in their journey to be completely carbon emission free. For now, they should rely on clean energy subsidies for enticing more and more people into adopting clean energy and adopting a hybrid model of energy production which will involve both clean energy and fossil fueled energy to be at pace with the growing requirements of clean energy without burning a hole in pockets of households or industries involved.

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